

Evolutionary System Prompt Optimization for LLM: Application to an Agent-Based Market Simulation

Abstract—Large Language Models (LLMs) are increasingly deployed as autonomous agents in interactive and multi-agent environments. In such settings, agent behavior is highly sensitive to system prompt design, yet prompt construction is typically manual and heuristic, making it difficult to optimize in the presence of stochastic interactions, delayed feedback, and non-differentiable objectives.

In this work, we introduce an evolutionary prompt-optimization framework for LLM-driven agents operating in competitive market environments. The framework treats system prompts as modular, discrete objects, effectively framing system prompt design as a combinatorial search problem over a discrete binary space, and applies a binary GA to evolve prompt configurations based solely on realized economic performance. This enables systematic optimization without requiring gradients, model fine-tuning, or access to internal model states.

We evaluate the proposed approach in an agent-based market simulation in which LLM-driven firms compete by setting prices and product quality under probabilistic consumer demand. Across a series of controlled experiments, the evolutionary process reliably identifies stable and interpretable prompt configurations, yielding substantial profit gains over both unoptimized prompt baselines and interpretable rule-based competitors. We further show that increasing stochasticity in LLM outputs degrades evolutionary convergence, and that optimized agent performance and prompt sparsity scale systematically with underlying model capability across multiple LLM families.

These results demonstrate that evolutionary prompt optimization provides a robust and model-agnostic mechanism for shaping LLM agent behavior in strategic environments, and reveal how model capability and prompt structure jointly influence emergent economic performance.

Impact Statement—LLMs are a popular technology for creating autonomous agents in interactive environments, given their ability to reason and adapt to contextual information during complex tasks. However, current research suggests that many agents perform suboptimally because their system prompts are constructed manually through trial-and-error.

The evolutionary prompt-optimization framework we introduce in this paper overcomes these limitations by applying a binary Genetic Algorithm to evolve modular prompt components based on realized performance. The purpose is to offer a robust and model-agnostic mechanism for shaping reliable AI behavior in complex, multi-agent strategic domains, even in a competitive setting.

Index Terms—large language models, LLM, genetic algorithms, prompt optimization, agent-based modeling, ABM, market simulations

I. INTRODUCTION

Large Language Models (LLMs) are increasingly deployed as decision-making agents in complex environments, ranging from recommendation systems and automated negotiation to multi-agent simulations and economic modeling [1]. Beyond their success in natural language tasks, recent work has shown that LLMs can exhibit non-trivial reasoning capabilities, adapt

to contextual information, and generate structured actions when embedded within interactive systems [2], [3], [4]. These properties have motivated their use as autonomous agents operating in dynamic, multi-agent settings.

Despite this growing interest, effectively controlling and shaping LLM behavior in strategic environments remains a fundamental challenge [5]. In practice, agent behavior is largely determined by prompt design, which specifies objectives, constraints, and contextual information [6]. Prompt construction, however, is typically performed in an ad hoc manner, relying on manual tuning, intuition, or trial-and-error [7], [8]. This approach becomes particularly problematic in multi-agent environments, where agent actions interact nonlinearly through competition and feedback, and where the mapping from prompt text to long-run outcomes is opaque, noisy, and non-differentiable [9], [10].

These challenges are especially pronounced in economic and market-based simulations [11]. In such settings, agents must repeatedly make interdependent decisions in uncertain scenarios, balance short-term gains against long-term profitability, and respond tactically and strategically to both competitors and consumers [12]. Small changes in agent policies can lead to large differences in aggregate outcomes, making systematic evaluation and optimization difficult. Agent-based modeling (ABM) is a computational approach in which complex system-level behavior emerges from the interactions of autonomous, heterogeneous agents following relatively simple decision rules [13]. While ABM provides a natural framework for studying such dynamics [14], integrating LLM-based agents into ABM raises new methodological questions regarding robustness, interpretability, and control.

In this work, we propose a novel evolutionary framework for optimizing LLM-driven agents, and we tested it in a competitive market environment where firms are modelled as autonomous agents that choose prices and product quality over repeated market periods, interacting with probabilistic consumers. Firm decision-making is implemented through structured system prompts provided to an LLM, and firm performance is evaluated in terms of cumulative economic profit. Rather than fixing prompts a priori, we optimize them using a binary Genetic Algorithm (GA) that evolves modular system prompt components based on available information and realized market performance.

Our approach addresses several core challenges inherent to LLM-based agent design. First, by restricting optimization to a discrete set of modular prompt components, we reduce the complexity of the prompt space while preserving a certain level of interpretability [15]. Second, the population-based nature of the GA allows exploration of a highly non-linear and non-differentiable objective landscape induced by multi-agent

interactions [16]. Third, by embedding optimization within a controlled agent-based market, we can directly measure economic outcomes and study the emergent strategic behavior.

Using this framework, we conduct a series of experiments designed to assess the methodological feasibility, analyzing the convergence properties of the GA under varying levels of stochasticity, compare optimized LLM-driven firms against unoptimized and rule-based baselines, and evaluate robustness across multiple LLM families and capability levels. To study how general model capability interacts with prompt optimization, we relate firm performance and prompt structure to publicly reported MMLU(-Pro) benchmark scores [17].

Results demonstrate that GA-driven prompt optimization converges reliably despite stochastic market dynamics and LLM variability, yielding stable and interpretable prompt configurations. Firms using optimized prompts achieve substantial and consistent profit advantages over both random-prompt baselines and hand-crafted rule-based competitors. Moreover, we find that model capability systematically affects both economic performance and the structure of optimized prompts: higher-capability models tend to require less explicit prompting to achieve strong strategic behavior, while architectural differences lead to distinct evolutionary trajectories.

This paper makes three main contributions. First, we introduce a general evolutionary framework for optimizing system prompts for LLM-driven agents operating in competitive, multi-agent environments. Second, we present a controlled agent-based market testbed that enables reproducible evaluation of LLM decision-making under economic competition. Finally, we provide empirical evidence linking LLM capability to both emergent economic performance and prompt complexity, highlighting systematic differences across model families.

Together, these findings advance the understanding of how LLM behavior can be shaped through evolutionary prompt optimization and provide a principled methodology for deploying LLM agents in strategic, multi-agent domains.

To support reproducibility, the complete source code for the simulation environment, the Genetic Algorithm implementation, and the final optimized prompt library are publicly available at: <https://anonymous.4open.science/r/llm-abm-optimization-CE70>.

II. BACKGROUND

This work builds on several strands of prior research, including LLMs as autonomous agents, prompt-based control, evolutionary optimization, and agent-based simulations of economic systems. In this section, we review the most relevant literature to situate our approach and motivate the methodological choices adopted in the remainder of the paper.

A. LLM Agents in Economic and Multi-Agent Environments

Recent advances in LLMs have led to their increasing use as autonomous agents embedded within interactive and multi-agent environments [1]. When provided with structured context and task-specific instructions, LLMs have been shown to exhibit non-trivial reasoning, planning, and decision-making

capabilities, enabling their deployment in simulations, negotiation tasks, and agent-based systems [2], [3], [18]. In such settings, the language model functions as a policy generator, mapping contextual descriptions of the environment to structured actions [19].

Economic and strategic environments pose particular challenges for LLM-driven agents [1]. Decisions are typically interdependent, unfold over multiple time steps, and must account for uncertainty, competition, and delayed feedback [20]. In these contexts, small variations in agent policies can lead to large differences in aggregate outcomes, making behavior difficult to predict or control [1]. Moreover, agent behavior has been shown to be highly sensitive to prompt formulation, instruction ordering, and contextual framing, underscoring the central role of system prompts in shaping emergent strategies [21], [22].

A further challenge is that LLM outputs are inherently stochastic, even under low-temperature decoding, introducing additional noise into long-horizon evaluations and complicating the assessment of agent performance in interactive environments [23], [24]. These properties motivate the need for robust and systematic methods for controlling and evaluating LLM-based agents in multi-agent economic settings.

B. Prompt Design as a Search Problem

Prompt design, often referred to as prompt engineering, has emerged as the primary mechanism for steering LLM behavior. Existing approaches typically rely on manually crafted instructions, few-shot examples, or stylistic constraints designed to elicit desired responses [25], [26]. While effective in many static or single-agent tasks, such methods are largely heuristic, labor-intensive, and difficult to generalize across environments [27], [28], [29].

In interactive and multi-agent settings, prompt design naturally gives rise to a search problem [30], [31]. The space of possible prompt configurations is combinatorial, high-dimensional, and non-differentiable, while the quality of a prompt can only be assessed indirectly through noisy, aggregate outcomes such as long-run performance or economic profit [32]. Human-driven optimization in this space is therefore brittle and non-systematic, motivating the use of automated search procedures capable of operating under uncertainty and delayed feedback [29], [33].

C. GAs for Discrete Optimization

GAs are a class of population-based evolutionary methods designed to optimize objective functions that are noisy, discontinuous, or non-differentiable [34]. By iteratively selecting, recombining, and mutating candidate solutions based on fitness, GAs can explore complex search spaces without requiring gradient information [34], [35].

These properties make GAs well-suited to discrete optimization problems with stochastic evaluation signals. In machine learning, evolutionary methods have been applied to hyperparameter tuning, architecture search, and policy optimization in settings where traditional gradient-based approaches are ineffective [36], [37]. Their robustness to noise and ability

to maintain population diversity are particularly advantageous in environments characterized by delayed rewards and non-stationary dynamics, such as those induced by interacting agents [35], [36].

D. Agent-Based Models and Market Simulations

ABMs provide a flexible framework for studying complex economic systems composed of heterogeneous, interacting agents [20], [14]. ABMs have been widely used to investigate market dynamics, competition, learning, and the emergence of macro-level phenomena from micro-level decision rules [38].

Classic ABMs rely on rule-based or hand-crafted agent behaviors, offering interpretability and analytical control but limited expressiveness [39], [20]. Beyond rule-based designs, a long tradition of agent-based models has employed neural networks—typically multilayer perceptrons—as cognitive engines for agent decision-making, enabling adaptive behavior while retaining a parametric structure [40]. Integrating LLM-driven agents into ABMs enables richer forms of decision-making and adaptation, but also introduces additional sources of stochasticity and non-linearity [3]. As a result, the resulting fitness landscape is typically irregular and non-differentiable [36], [37].

Rule-based agents remain a common modeling choice in agent-based economic simulations due to their transparency, interpretability, and reproducibility. Such agents provide stable and easily understood baselines against which more complex decision-making mechanisms can be evaluated [41], [42].

In the context of LLM-driven agents, rule-based firms are commonly used as transparent and reproducible baselines in agent-based economic simulations [20], [39]. Comparing LLM-driven agents against such baselines provides a reference point for assessing performance differences in complex and stochastic environments [38].

To compare agent behavior across different LLMs, prior work often relies on external benchmark scores as coarse proxies for general model capability [43]. While such measures do not directly capture strategic competence, they provide a useful reference for analyzing how underlying model capacity interacts with prompt structure and emergent behavior in multi-agent environments [28].

Together, these strands of research motivate a framework that combines agent-based economic simulations, evolutionary optimization, and LLM-driven decision-making.

III. METHODOLOGY

A. Agent-based model

This section outlines the methodology employed to simulate firm and consumer behavior within a competitive market environment. The model stylizes the dynamic interactions between autonomous firms and consumers [13], [42], where the agents make decisions based on past performance and market conditions. The simulation is designed to capture the changing nature of market dynamics [38] by allowing firms to adjust two decision variables, which are price and quality, over time in response to both competitor actions and consumer behavior [14]. The market operates in discrete time steps, and

at each step, agents interact with one another in a probabilistic environment.

The following subsections examine the roles of the agents within the simulation, starting with the firm agents, followed by the consumer agents, and concluding with the broader market interaction loop.

1) *Firm Agents*: Firms are modeled as autonomous economic agents operating in a competitive market environment. At each market period, every firm independently selects a selling price and a product quality, with the objective of maximizing cumulative profit over time. The firm's primary objective is to maximize its cumulative profit over a sequence of market periods by strategically selecting a selling price (p) and a product quality (q) at each decision point. These strategic choices determine consumer demand, realized sales, and ultimately firm success in the market.

Firms interact repeatedly and post their price–quality offer to the market at each time step, compete for consumers, consequently receiving a feedback in the form of realized sales and profits. This feedback informs subsequent decisions, allowing firms to adapt their behavior over time in response to both market conditions and competitor actions.

At the end of each period, firms update their cumulative profit based on realized market outcomes. Given a selling price p , a chosen quality q , and the number of units sold u , the profit increment is defined as:

$$\Delta\pi = u(p - c), \quad (1)$$

where c denotes the unit cost of production. Production costs are modeled as linearly proportional to product quality:

$$c = kq, \quad (2)$$

with k representing the cost index parameter used in the simulations, in all experiments reported in this paper, we fix $k = 0.75$. Cumulative profit at time t is updated recursively as:

$$\pi_{t+1} = \pi_t + \Delta\pi. \quad (3)$$

The simulation includes two distinct types of firm agents. The first, and primary focus of this work, consists of LLM-driven firms, whose decisions are generated by querying a LLM using a structured system prompt (details are shown later in the experimental design section). The second type consists of rule-based firms, which follow a fixed and fully interpretable heuristic. These rule-based agents are not intended to model sophisticated strategic reasoning; instead, they serve as a baseline that allows us to benchmark the performance of LLM-driven firms against a non-learning decision mechanism.

a) *LLM-driven Firms*: Each LLM-driven firm is an autonomous decision-making agent powered by a LLM. The decision-making process unfolds in two stages. At the beginning of each period, the firm constructs a structured system prompt that summarizes relevant market information, including cost structure, observed competitor conditions, and aggregate consumer tendencies. This prompt is provided to the language model, which responds with a structured output of the form `DECISION: (p, q)`, where $p \in [0.1, 1]$ and $q \in [0, 1]$. The firm then adopts these values as its strategy for the current period.

b) *Rule-Based Firms*: We additionally include a class of rule-based firms in the simulation. Their role is methodological rather than descriptive: they provide a non-learning reference point that enables clean and reproducible comparisons, isolating the economic advantages arising from LLM-based decision-making.

Because the behavior of rule-based firms must be fixed and transparent, performance differences can be attributed directly to the strategic flexibility induced by the LLM prompts, rather than to learning dynamics or hidden algorithmic complexity in the rule-based agents. Accordingly, we intentionally adopt a lightweight heuristic design instead of a sophisticated optimization strategy.

Each rule-based firm follows a fixed, non-learning heuristic that adjusts its price and quality solely in response to realized profit in the previous period. Firm behavior consists of two stages: an initialization step and a profit-reactive update rule.

Initialization.: At period $t = 0$, each firm draws an initial product quality

$$q_0 \sim U(q_{\min}, q_{\max}),$$

where in the experimental setting $q_{\min} = 0.3$ and $q_{\max} = 0.7$, and sets its initial price using a cost-based markup:

$$p_0 = k q_0 + m_0, \quad (4)$$

where m_0 is also sampled from an uniform distribution, ensuring positive expected margins and heterogeneous starting conditions across agents. The unit cost is then $c_0 = k q_0$.

Update Rule.: For each period $t > 0$, the rule-based firm observes its realized profit from the previous period, π_t , and updates price and quality using a simple profit-reactive heuristic with bounded random exploration. Let $\epsilon > 0$ denote a small tolerance for “approximately zero” profit, and let $\varepsilon_p \sim U(-\eta_p, \eta_p)$ and $\varepsilon_q \sim U(-\eta_q, \eta_q)$ be independent noise terms. The firm then applies:

$$p_{t+1} = \begin{cases} p_t + \Delta p_{\text{loss}} + \varepsilon_p, & \pi_t < -\epsilon, \\ p_t + \Delta p_{\text{zero}} + \varepsilon_p, & |\pi_t| \leq \epsilon, \\ p_t + \varepsilon_p, & \pi_t > \epsilon. \end{cases} \quad (5)$$

$$q_{t+1} = \begin{cases} q_t - \Delta q_{\text{loss}} - \varepsilon_q, & \pi_t < -\epsilon, \\ q_t + \Delta q_{\text{zero}} + \varepsilon_q, & |\pi_t| \leq \epsilon, \\ q_t + \varepsilon_q, & \pi_t > \epsilon. \end{cases} \quad (6)$$

Finally, the updated values are clipped to the admissible ranges $p_t \in [0.1, 1]$ and $q_t \in [0, 1]$, and unit cost is updated as $c_t = k q_t$. For $t = 0$, the firm uses its initialized (p_0, q_0) without applying the update rule.

Specific numerical values and parameter ranges used in the experiments are reported in the Experimental Design section.

A schematic flow of the update mechanism is shown in Fig. 1.

2) *Consumer Agents*: Consumers are modeled as rational but stochastic agents who seek to maximize their utility [44]. The utility function with respect to firm j captures a preference for higher-quality products at lower prices and is defined as:

$$U_j = \frac{q_j}{p_j} \quad (7)$$

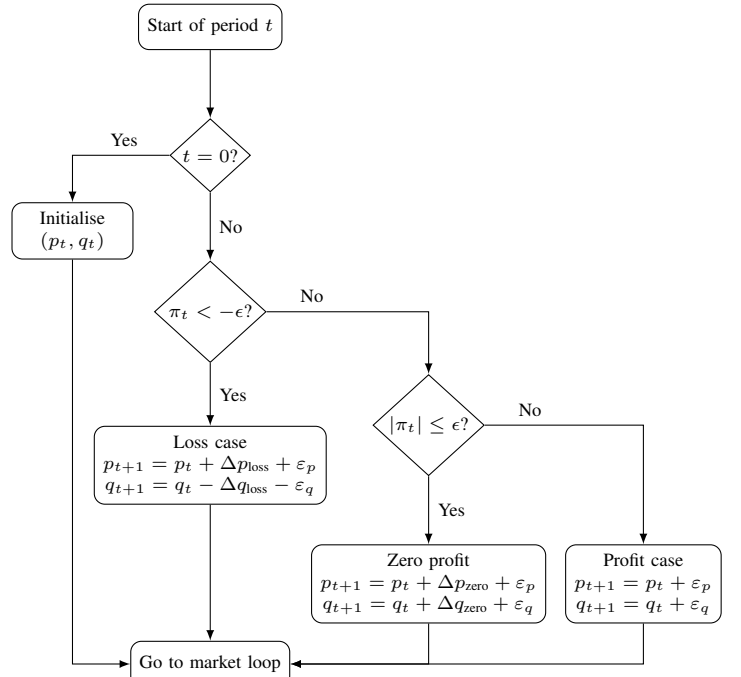


Fig. 1. Schematic decision flow of the rule-based firm.

where p_j and q_j denote the price and quality, respectively, of the product offered by firm j . It is a stylized model that takes into consideration the limitations of real-world decision-making [45], we incorporate the concept of bounded rationality by assuming that consumers do not always select the firm offering the highest utility [44]. Instead, their choices are modeled probabilistically to account for market noise and individual heterogeneity.

To model this, we employ a softmax function that transforms utility scores into a probability distribution [46]. The probability P_{ij} that consumer i selects firm j is given by:

$$P_{ij} = \frac{e^{U_j/\tau}}{\sum_{l=1}^n e^{U_l/\tau}} \quad (8)$$

Here, n denotes the total number of firms participating in the market simulation, ensuring that the probabilities are normalized over all available choices. We introduce a temperature parameter τ to control the degree of randomness in consumer behavior. Lower values of τ yield more deterministic choices, favoring firms with higher utility, whereas higher values lead to more exploratory behavior, allowing consumers to consider a wider range of options. This formulation captures both the stochasticity of individual preferences and the imperfect nature of real-world decision-making.

3) *Market Interaction Loop*: These agents interact within a structured market loop over a predefined number of periods, where each time step in the simulation follows a structured four-phase interaction loop, schematically illustrated in Fig. 2:

a) *Firm Decision*: Each firm receives structured market information, which is then processed by a LLM to generate optimal price and quality settings. These outputs are adopted by the firm for the current period.

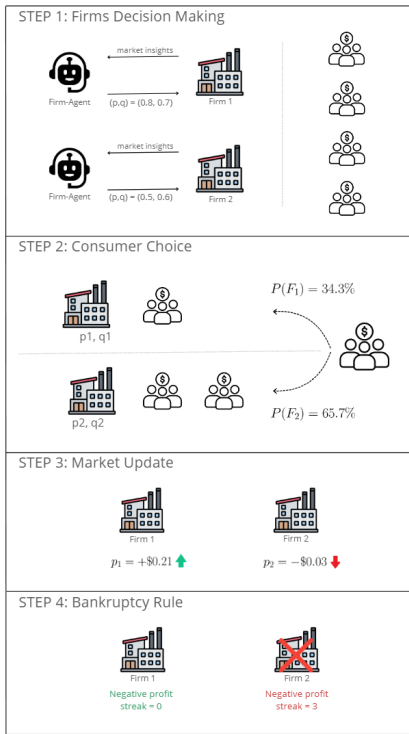


Fig. 2. Schematic representation of a single time step in the market loop simulation, illustrating two LLM-driven firms and four buyers across the four core phases: (1) firm decision-making, (2) consumer choice, (3) market update, and (4) bankruptcy rule.

- b) **Consumer Choice**: Each consumer purchases exactly one unit per period. Consumers evaluate the available firm offerings and select a single firm probabilistically based on the quality-to-price ratio of each product.
- c) **Market Update**: Firms receive feedback from the market in the form of sales outcomes. Key performance indicators—such as profits, revenues, and units sold—are updated accordingly.
- d) **Bankruptcy rule**: If a firm records three consecutive periods of negative profit, it is marked as bankrupt and immediately exits the market. Once removed, the firm no longer participates in subsequent decision or competition phases.

All prices, costs, and profits are expressed in arbitrary simulation units that are known to the LLMs; since each consumer buys one unit per period, aggregate profits scale linearly with the number of consumers and the number of periods.

B. LLM Prompting Interface: Models and Prompting

The decision layer of each LLM-driven firm is implemented as a structured prompting interface. At every period, the agent receives a textual description of the market state and produces a pair of continuous actions (p, q) , which are then executed in the simulation. In this subsection we describe the prompt templates used to query the models and the set of language models evaluated, together with the capability measure adopted for cross-model analyses.

1) *Prompt Templates*: Each LLM-controlled firm is queried using a composite prompt consisting of a system prompt and a minimal user prompt. The system prompt is the only component that carries task-relevant information and is the target of optimization. This restriction is a deliberate modeling choice that allows us to study prompt optimization in a simplified and controlled setting, reducing search-space complexity and minimizing interference effects between prompt components. Since both system and user prompts ultimately shape agent behavior, the insights derived from this controlled setting are expected to generalize to richer prompt configurations, while providing a clearer interpretation of the underlying mechanisms.

The system prompt itself is constructed by concatenating a set of short, modular components, each encoding a specific instruction or piece of contextual information (e.g., profit objective, cost structure, competitive setting), for a total of eight candidate components in the baseline configuration. A binary genome specifies which of these components are included in a given system prompt, as described in Section III-D. The user prompt is kept fixed across all experiments and does not introduce any additional market-specific information; it merely requests that the model output its decision in the required structured format (e.g., `DECISION: (price = ..., quality = ...)`).

2) *LLM Model Suite and Capability Measure*: To assess the generality of the proposed framework and to investigate how model capability interacts with prompt optimization, we evaluate a diverse suite of LLMs spanning multiple capability levels and architectures. Specifically, we consider four GPT models—GPT-3.5-turbo, GPT-4o-mini, GPT-5-mini, and GPT-5-nano—and three Gemini models—Gemini-2.0-flash, Gemini-2.5-flash and Gemini-3-flash-preview. These models were selected to cover a broad range of publicly reported performance on general-purpose benchmarks, while including both intra-family and cross-family comparisons.

All models share the same interaction protocol: they are queried with the structured prompt described above and are required to emit a price-quality pair in the specified output format. For models exposing a temperature parameter (e.g., GPT-3.5-turbo, GPT-4o-mini, Gemini-2.0-flash, Gemini-2.5-flash, Gemini-3-flash-preview), decoding is performed with $T = 0$ in all main experiments to reduce sampling noise and ensure deterministic responses for a given prompt. Models that do not expose temperature control (notably GPT-5-mini and GPT-5-nano) are used with their default sampling behavior, which corresponds to a higher effective temperature.

To ensure that the observed performance is strictly a function of the optimized system prompt rather than the model's internal Chain-of-Thought (CoT), we strictly constrained the generation parameters. All model outputs were capped at a maximum of 30 tokens, neutralizing implicit multi-step reasoning [47], [48]. Furthermore, for reasoning-capable models (such as the Gemini-3-flash-preview, GPT-5-mini, and GPT-5-nano), the internal thinking budget was explicitly set to zero to force immediate, zero-shot structured decisions.

As an external measure of model capability, we use publicly reported scores on the MMLU(-Pro) benchmark [49], [17]. For

each model, we record (i) the cumulative profit achieved by the firm using its GA-optimized system prompt and (ii) the bit-density of the resulting prompt, measured in terms of active components.

C. GA for System Prompt Optimization

Rather than relying on fixed heuristics or predefined decision rules, firm behavior emerges from the design of the system and of user prompts provided to the LLM. As such, improvements in firm performance are directly related to the quality of these prompts. Well-designed prompts can guide the LLM toward economic rational behavior, while poorly constructed ones may lead to suboptimal or erratic strategies.

However, optimizing prompts presents several unique challenges:

- **High-dimensional search space:** Prompts may contain many configurable components, such as goals, constraints, instructions, and tone, resulting in a vast combinatorial design space.
- **Non-differentiable objective:** The performance metric (e.g., cumulative profit) cannot be expressed as a smooth or differentiable function of the prompt text, making gradient-based optimization infeasible.
- **Local optima:** The space of prompt configurations is irregular and discontinuous, often containing many local optima that can mislead greedy or hill-climbing search strategies.

To address these challenges, we employ a heuristic optimization method—specifically, a GA [50], [51] to evolve system prompts over successive generations. GAs are well-suited to this task due to their population-based search, ability to explore multiple regions of the solution space simultaneously, and robustness against non-linearity and noise [15], [52]. In our framework, the GA iteratively refines a population of candidate prompts, evaluating their fitness based on the economic performance of the firm agents that use them.

This evolutionary approach enables us to discover prompt configurations that yield significantly improved firm outcomes in competitive and stochastic market environments.

1) *Binary GA Structure:* Given the challenges of optimizing LLM prompts, we designed a custom binary GA architecture tailored to this task. To reduce complexity, the GA is applied only to the system prompt, while the user prompt template is kept fixed, operating on a population of binary-encoded system prompts and evolves them over successive generations to improve firm performance in the market simulation. We adopt a co-evolutionary evaluation scheme in which the GA population size coincides with the number of firms in the market: each genome in the population is instantiated as one firm, and fitness is evaluated through simultaneous competition against all other individuals in the population. The main components of our architecture are as follows.

a) *Encoding Strategy:* System prompts are constructed from a library of 8-14 modular components, each representing a specific instruction, constraint, or environmental fact (e.g., market context, firm objectives, strategic advice). Each prompt

Algorithm 1 Binary GA for System Prompt Optimization

Require: System component pool $C = \{c_1, \dots, c_n\}$; population size N ; generations T ;

tournament size k ; elitism E ; mutation prob. p_{mut} ;
fitness $f(\cdot)$ (cumulative profit over T_{sim} periods)

- 1: Initialize $P_0 = \{g_1, \dots, g_N\}$, each $g_i \in \{0, 1\}^n \triangleright$ system bits only
 - 2: **for** $t = 1$ to T **do**
 - 3: Decode each genome $g_i \in P_t$ into a system prompt p_i
 - 4: Run one market simulation with N firms, assigning prompt p_i to firm i
 - 5: Obtain fitness scores $(s_1, \dots, s_N)$ where s_i is firm i 's cumulative profit
 - 6: E elites $\leftarrow$ top- E genomes by s_i
 - 7: Select remaining parents via k -tournament on $\{s_i\}$
 - 8: Apply crossover, mutation, form P_{t+1}
 - 9: **end for**
 - 10: **Return:** Best prompt $p^* = \arg \max_{g \in P_T} f(\text{decode}(g))$
-

is encoded as a fixed-length binary string, where each bit corresponds to the inclusion (1) or exclusion (0) of a component. Thus, a genome of length n defines a system prompt with n potential segments, enabling a combinatorially rich search space of 2^n possible configurations.

b) *Population Initialization:* The initial population consists of N binary genomes $g_i \in \{0, 1\}^n$ sampled uniformly at random, each encoding a candidate system prompt via the inclusion or exclusion of the n available system components.

c) *Fitness Evaluation:* Each binary genome is decoded into a complete system prompt and assigned to a firm agent. These firms then participate in a full market simulation, where the different prompt variations compete against one another. The fitness of a genome is defined as the cumulative profit generated by the corresponding firm over all simulation time steps. Formally, the fitness function f for each firm is given by:

$$f = \sum_{t=1}^T u_t \cdot (p_t - k \cdot q_t) \quad (9)$$

where T denotes the total number of market periods, p_t and q_t are the price and quality chosen by the firm at period t , and production costs are linear in quality with unit cost $c_t = kq_t$, where $k > 0$ is the cost index parameter. The variable u_t represents the number of units sold at period t . In our market model, each consumer purchases exactly one unit per period; therefore, u_t is equal to the number of consumers who select the firm at time t under the probabilistic choice rule.

d) *Selection Mechanism:* Parent selection is performed using tournament selection, which ensures that higher-performing individuals have a greater chance of reproduction while maintaining population diversity. This mechanism provides a balance between exploitation of good solutions and exploration of the broader search space, as discussed in standard GA frameworks [35]. Crossover is applied with a fixed probability using single-point crossover. A random crossover point is selected, and two parent genomes exchange

segments to produce two offspring [35]. To maintain genetic diversity and avoid premature convergence, mutations are also applied to each offspring by flipping individual bits with a small probability p_{mut} . Each generation entirely replaces the previous population. Optionally, elitism is employed to carry over the top-performing individual unchanged, preserving the best solution found so far and ensuring that the quality of solutions does not degrade across generations [35]. The algorithm terminates after a fixed number of generations or when the best individual’s genome plateaus for a predefined number of iterations, to ensure computational efficiency while allowing sufficient time for convergence.

This constitutes the general architecture of our GA for prompt optimization. Because fitness is evaluated in a shared competitive environment, it is inherently relative and depends on the contemporaneous strategies present in the population. This coupling is intentional: it mirrors real economic competition and ensures that prompts are selected for robustness against a diverse and evolving set of opponents, rather than against a fixed baseline.

However, due to the high degree of stochasticity introduced by LLM variability and probabilistic consumer behavior [53], additional techniques were necessary to ensure robust and reliable evaluations. These are detailed in the next subsection.

D. Experimental Design

To account for potential API updates and model drift in proprietary LLMs, all simulations and data collection reported in this study were conducted between September 2025 and December 2025. This section describes the experimental protocols used to evaluate the proposed prompt-optimization framework, designed to enable a binary GA to reliably converge to stable and interpretable system prompt components in a noisy ABM–LLM environment, and to identify the effect of LLM stochasticity (e.g., sampling temperature) on GA convergence. We also aim to confirm that GA-optimized prompts yield measurable competitive advantages over unoptimized and rule-based firms, that the evolutionary framework generalizes across LLM families, and to examine how model capability relates to economic performance and prompt complexity.

To answer these questions, we conduct a set of controlled experiments using the default configuration reported in Table I. All GA runs use binary-encoded system prompts and identical evaluation procedures unless stated otherwise. Each configuration is evaluated over multiple independent GA runs, providing robust estimates of convergence behavior and performance.

In addition to the global GA and market configuration, rule-based firms are instantiated using a fixed set of numerical parameters that control their initialization, adjustment steps, and bounded exploration. These values are held constant across all experiments to ensure comparability and reproducibility. The full parameterization of the rule-based agents is reported in Table II.

A relatively large number of consumers (on the order of 10^5) is used to reduce sampling noise in demand allocation and thereby obtain a smoother and more informative fitness signal for the GA. All pseudorandom number generators (Python

TABLE I
GA AND MARKET CONFIGURATION USED IN THE MAIN EXPERIMENTS

Parameter	Value
Population size	50
Generations	15–30
Tournament size	6
Elitism	5
Mutation rate	0.10
Buyers per market	80,000–100,000
Periods per market	10
Cost index k	0.75
Consumer temperature τ	0.10 (fixed)
LLM temperature	0 (unless varied)

TABLE II
RULE-BASED FIRM PARAMETERS.

Parameter	Description	Value
$[p_{\min}, p_{\max}]$	Price bounds	$[0.1, 1.0]$
$[q_{\min}, q_{\max}]$	Quality bounds	$[0.0, 1.0]$
$q_0 \sim U(\cdot)$	Initial quality range	$U(0.3, 0.7)$
$m_0 \sim U(\cdot)$	Initial margin range	$U(0.05, 0.2)$
Δp_{loss}	Loss price step	+0.05
Δp_{zero}	Zero-profit price step	−0.02
Δq_{loss}	Quality step (loss)	0.05
Δq_{zero}	Quality step (zero profit)	0.05
η_p	Price noise half-range	0.01
η_q	Quality noise half-range	0.01
ϵ	Zero-profit tolerance	10^{-6}

3.11.9 and NumPy 1.26.4) are seeded for reproducibility; different seeds are used across independent runs, while each individual run is deterministic conditional on its random seed. Each fitness evaluation consists of one full market simulation, and the cumulative profit of the corresponding firm is used as the fitness score. All reported metrics aggregate results across the independent GA runs associated with each experimental condition.

A key methodological challenge in this setting concerns the interaction between GA hyperparameters and the underlying language model. In preliminary experiments, GA configurations that exhibited stable convergence when tuned on a single reference model often failed to converge or produced unstable evolutionary dynamics when applied unchanged to other LLMs. The parameter set reported in Table I reflects the outcome of an empirical calibration process aimed at identifying a unified configuration that supports stable GA behavior across all evaluated models. All experiments reported in this paper adopt this shared configuration, ensuring cross-model comparability without introducing model-specific adjustments.

1) Convergence Experiments:

a) *Baseline convergence in an 8-component search space:* We first assess the GA’s ability to identify high-utility prompt components in a compact discrete landscape. In the baseline setting, system prompts are constructed from a pool of eight empirically beneficial components (e.g., explicit profit-maximization goals, cost-awareness, competitive reasoning). Each prompt is encoded as an 8-bit genome, corresponding to $2^8 = 256$ possible configurations, where each bit indicates the inclusion or exclusion of a specific component. We per-

form multiple independent GA runs under identical market conditions and record: (i) the genotype of the best individual per generation and (ii) population-level activation frequencies of all components. This experiment evaluates the strength of selection pressure and the consistency of convergence in a relatively small, well-structured search space.

b) Robustness in an augmented 14-component search space: To test robustness in a larger and more heterogeneous search space, we expand the genome to 14 bits by adding six additional components to the original pool. These new components fall into three categories: (i) *neutral* phrases, which are semantically redundant or operationally irrelevant and are expected to have negligible impact on behavior; (ii) *mildly adversarial* phrases, which induce small but systematic inefficiencies (e.g., encouraging excessively high quality or prioritizing fairness over profit); and (iii) *strongly adversarial* phrases, which directly undermine performance (e.g., instructing the firm to minimize profit or offer products for free). All components are written in a comparable style and length to avoid trivial detection based on surface form alone. GA and market hyperparameters are kept identical to the 8-gene experiments. By tracking best-individual genotypes and activation frequencies across generations and runs, we examine whether the GA suppresses harmful instructions, remains insensitive to neutral ones, and converges reliably despite the increased dimensionality and semantic heterogeneity of the search space.

2) Control of Consumer-Level Stochasticity: To ensure that the fitness landscape remained interpretable, we fixed the softmax temperature of the consumer choice model to a low value ($\tau = 0.1$) in all experiments. Higher values of τ would introduce additional randomness in demand allocation, increasing variance in firm profits and confounding the effect of LLM-induced stochasticity on GA convergence. Since our primary goal is to isolate the impact of LLM sampling temperature, buyer stochasticity is treated as a controlled variable rather than an experimental dimension. We therefore do not conduct explicit τ -sweeps.

3) Sensitivity to LLM Stochasticity: To evaluate how noise originating from the LLM affects the evolutionary dynamics, we vary the LLM sampling temperature over

$$T \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$$

a range that spans near-deterministic decoding ($T = 0$) to highly stochastic behavior ($T = 1$). Ten independent GA runs are executed at each temperature level. In all cases, market settings, GA hyperparameters, and consumer stochasticity are held strictly constant to ensure that differences in convergence behavior can be attributed solely to the LLM temperature.

Each run produces a “best individual” (i.e., the highest-fitness genotype at termination). The empirical distribution of these genotypes across the ten runs at each temperature level serves as the key measurement of convergence stability. These results are analyzed in Section IV-B.

4) Market-Level Performance Experiments:

a) Optimized vs. unoptimized baseline firms: To measure the economic advantage conferred by optimized prompts,

we construct markets with one GA-optimized firm and nine baseline firms. Baseline firms are initialized with randomly sampled, fixed genomes. Each market is simulated over 10 periods and repeated across 20 independent trials with newly resampled baseline competitors. We record cumulative profit for all firms to quantify performance differentials attributable solely to prompt optimization.

b) Optimized LLM firms vs. rule-based firms: Before evaluating LLM-driven firms against rule-based competitors, we first validated that the rule-based agents constitute a meaningful and economic coherent baseline. To this end, we ran a stress-test market containing 100 independently initialized rule-based firms and 100,000 consumers over 10 periods. This diagnostic experiment verifies that rule-based agents are capable of generating positive cumulative profit under realistic conditions, their behavior remains heterogeneous, producing a non-degenerate competitive landscape, and the market environment is well-calibrated and free of pathological dynamics. The resulting distribution of cumulative profits is reported in Section IV.

To compare LLM-driven policies with hand-crafted heuristics, we run markets containing one optimized LLM agent and five rule-based firms; each simulation is repeated five times under identical conditions. This setup provides a clean comparison between rule-based baselines and LLM-driven decision-making.

5) Cross-Model Generalization and Capability Analysis: To assess generalization across architectures, we replicate the full prompt-optimization pipeline for seven models from two different families: GPT-3.5-turbo, GPT-4o-mini, GPT-5-mini, GPT-5-nano, Gemini-2.0-flash, Gemini-2.5-flash, and Gemini-3-flash-preview. For each model, we perform three independent GA runs using identical hyperparameters. During each run, we record profit trajectories, mean active-bit density per generation, and final optimized genomes. Publicly reported MMLU(-Pro) scores are used as external capability estimates. After optimization, we compute the correlation between each model’s MMLU score and the cumulative profit of its optimized agent and the compactness of its final system prompt. This analysis enables us to study how model capability relates to economic performance and prompt-complexity requirements.

IV. RESULTS AND DISCUSSION

This section evaluates (i) the feasibility of using a binary GA to optimize system prompts for LLM-driven firms, (ii) the performance gains achieved by optimized prompts in competitive markets, (iii) the sensitivity of GA convergence to stochasticity in the simulation, and (iv) the relationship between model capability and the performance and prompt-structure of the resulting optimized agents.

A. GA Feasibility and Convergence

We first assess whether a binary GA can reliably identify high-utility prompt components in a discrete, non-differentiable, and noisy objective landscape. All experiments

in this subsection were conducted under identical market and GA hyperparameters (Table I), using both the baseline search space of eight components and an augmented space of fourteen components as described in the experimental design.

With the baseline search space (8 components), the GA of system prompt components exhibits rapid and highly consistent convergence. As shown in Fig. 3 and Fig. 4 (respectively best individual per generation and population activation frequencies), the algorithm concentrates on a stable subset of beneficial components within the first five generations. Once this configuration emerges, it remains the dominant solution for the remainder of the run, indicating a clear selection gradient despite the stochasticity of market demand. Low-impact components are progressively eliminated from the population, and the overall population structure becomes increasingly homogeneous around the optimal genotype.

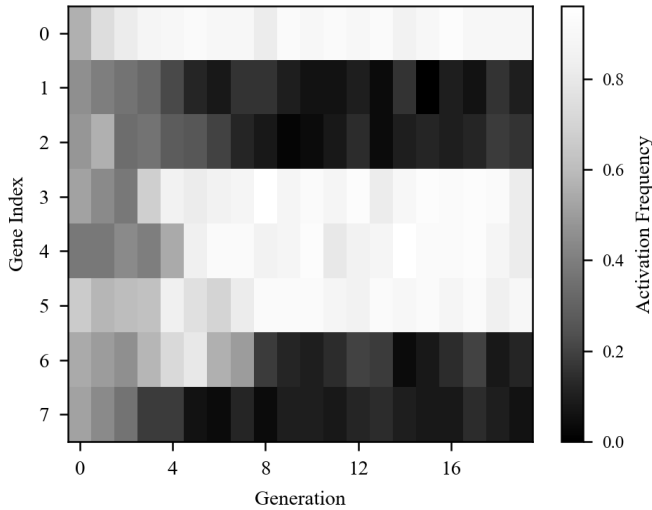


Fig. 3. Population-level activation frequencies for the 8-gene search space. Colors indicate the fraction of individuals activating each component across generations.

To evaluate robustness, we expanded the search space from $2^8 = 256$ to $2^{14} = 16,384$ possible genotypes by further adding six additional components [51] as specified before. Figures 6 and 5 report the corresponding convergence dynamics.

Also in this case, the GA of system prompt components continues to display strong and interpretable selection behavior. Adversarial components are rapidly suppressed: their activation decreases from an initial $\sim 50\%$ (random initialization) to below 10% by the third generation, after which they persist only at levels consistent with the mutation rate (10%), indicating complete elimination through selection. Mildly adversarial components follow a similar pattern, reaching $\sim 8\%$ activation by the fifth generation and remaining at mutation-level frequencies thereafter.

Neutral components, which have been engineered to have negligible influence on firm performance, show a different

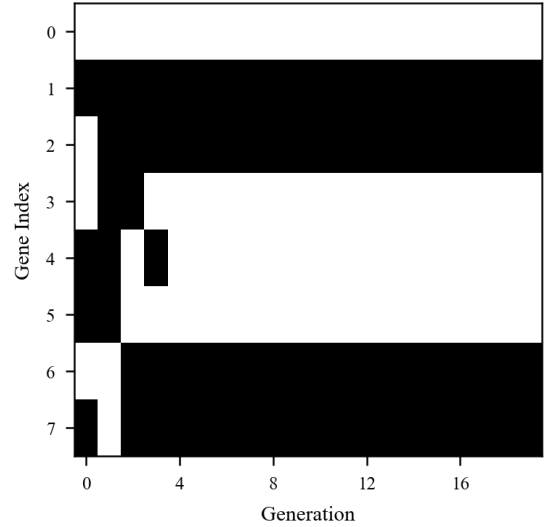


Fig. 4. Gene activation in the best individual across generations (8-gene search space). White cells mark components active in the highest-performing genotype at each generation.

trajectory. Their activation initially increases, reaching a peak of approximately 85% around generation 10. This reflects the GA’s exploratory dynamics: in the absence of selection pressure, neutral instructions drift freely within the population and may appear in high-performing individuals during intermediate generations [35]. However, once the global optimum stabilized, their activation declines to $\sim 10\%$ in the final generations, again matching the mutation floor. This also support the experimental decision regarding the number of generations. Consistently across runs, the final best-individual genotype in the 14-gene experiment matches exactly the optimum found in the baseline 8-gene space, with the additional components switched off, confirming the goodness of the methodology and the fact that the selection process is consistent.

The increased search-space size modestly delays convergence: the optimal genotype first appears around generation 12 in the 14-gene run, compared to generation 5 in the 8-gene baseline. Nevertheless, once identified, the optimal genotype remains stable and dominates the population through the final generation in both settings. These results confirm that the GA is able to detect and reinforce beneficial prompt components, suppress detrimental instructions even under substantial noise, and also remain insensitive to semantically neutral content. Overall, the convergence patterns demonstrate that the GA yields a stable, interpretable, and repeatable set of prompt components despite the discrete search space and stochastic firm–market interactions.

B. Sensitivity to LLM Stochasticity: Effect of Temperature on GA Convergence

To systematically assess how stochasticity in LLM outputs affects the GA’s convergence, we conducted a controlled experiment varying the sampling temperature parameter of the

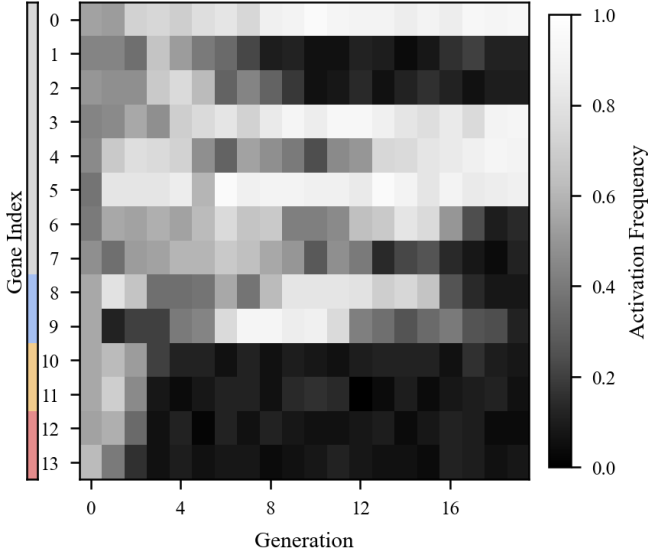


Fig. 5. Population-level activation frequencies in the augmented 14-gene search space. Population-level activation frequencies in the augmented 14-gene search space. Additional components (indices 8–13) include neutral (blue), mildly adversarial (orange), and adversarial (red) phrases.

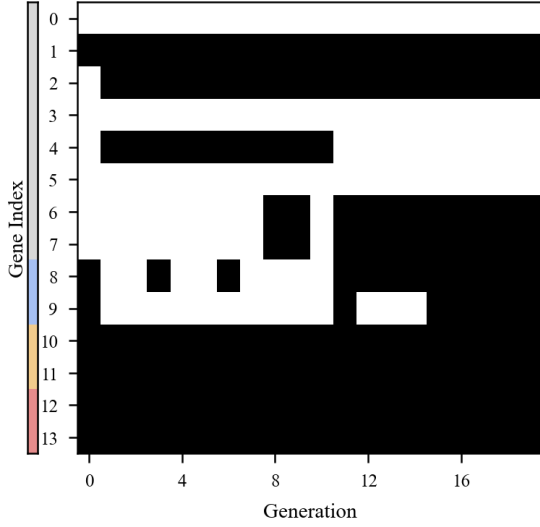


Fig. 6. Gene activation in the best individual across generations (14-gene search space). Additional components (indices 8–13) include neutral (blue), mildly adversarial (orange), and adversarial (red) phrases.

underlying language model. Temperature values ranged from 0.0 to 1.0 in increments of 0.2, with ten independent GA runs executed at each level under otherwise identical simulation settings.

Figure 7 reports the frequency with which each genotype (i.e., the binary representation of the best individual per run) appeared as the optimal solution across temperatures. The results reveal a clear pattern: as temperature increases, the GA’s convergence becomes less stable and more dispersed across alternative genotypes. At a temperature of 0, the algorithm consistently converges to the same optimal solution, indicating strong selection pressure and a well-defined fitness

landscape. As temperature rises, stochastic variations in the LLM responses inject noise into the evaluation process, effectively distorting the fitness signal perceived by the GA and preventing consistent convergence.

An alternative interpretation is that varying the temperature may also reshape the underlying decision landscape: the mapping from prompts to firm actions, and hence to profits, could change with temperature, shifting the position of the global optimum itself. Nevertheless, the persistent lack of convergence to any unique solution at higher temperatures suggests that noise, rather than a systematic shift in the optimum, is the dominant factor. These results reinforce the methodological choice of adopting deterministic LLM settings (temperature = 0) whenever possible, as discussed in Section III-D.

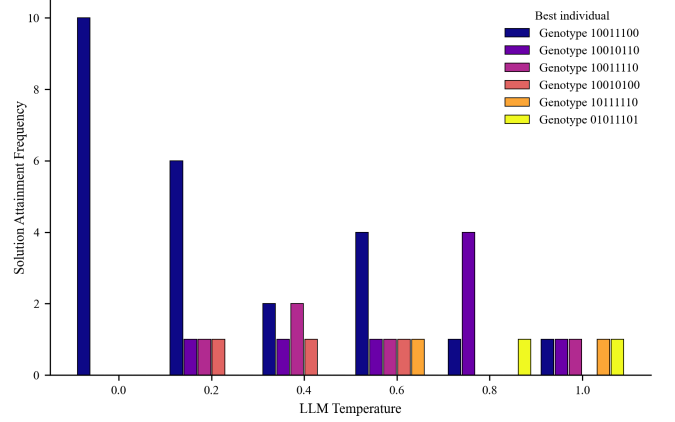


Fig. 7. Effect of LLM temperature on GA convergence stability (GPT-4o-mini). The bars report the frequency with which each genotype appeared as the best individual across ten runs per temperature level. Lower temperatures yield consistent convergence toward a single optimal prompt, while higher temperatures produce more diverse and unstable outcomes.

C. Market-Level Performance Gains From Prompt Optimization

To quantify the economic effect of prompt optimization, we evaluate firms in open competition. Each market contains one firm using the GA-optimized system prompt and nine baseline firms initialized with fixed, randomly generated genomes. Each baseline prompt corresponds to a distinct bit-string drawn uniformly at random at the start of the experiment and held constant throughout the simulation. This design ensures that baseline performance reflects the profitability of unoptimized prompt configurations rather than stochastic variation within a single prompt.

For each configuration, we simulate ten trading periods and repeat the experiment across twenty independent markets with newly resampled baseline competitors. Figure 8 reports the cumulative profit obtained by the optimized firm compared to the mean performance of baseline firms. The optimized firm consistently achieves substantially higher profits, with an average cumulative profit of approximately 14,700 units, whereas baseline firms incur an average loss of roughly 8,500 units.

These results demonstrate that GA-driven prompt optimization yields not only a stable and interpretable set of system prompt components, but also confers an observable competitive advantage in multi-agent market environments.

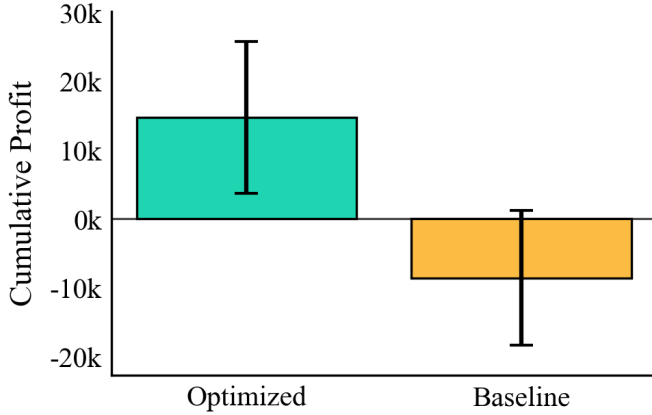


Fig. 8. Cumulative profit of the GA-optimized firm versus the average baseline firm across 20 independent markets (mean $\pm$ standard deviation). The optimized prompt consistently outperforms non-optimized prompts in open competition.

D. LLM-Optimized Agents vs. Rule-Based Firms

To compare optimized LLM agents to heuristic competitors, we first verified that the rule-based firms constitute a meaningful and economic viable baseline. Figure 9 reports the distribution of cumulative profits generated by 100 rule-based firms competing against one another for 100,000 consumers over 10 periods. The broad, positively supported distribution confirms that rule-based agents reliably generate profit and display heterogeneous competitive behavior, ensuring a suitable reference environment.

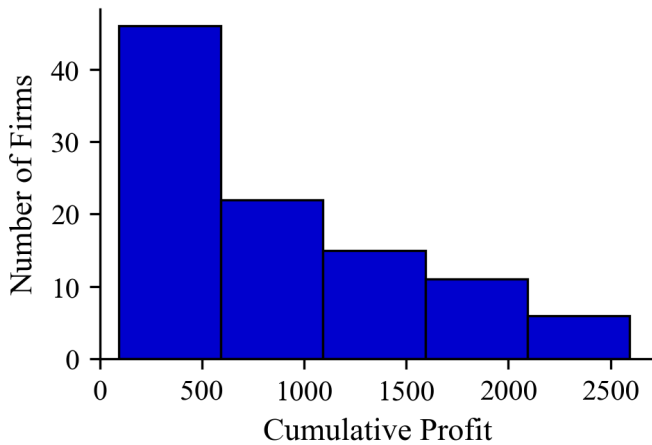


Fig. 9. Distribution of cumulative profits for 100 rule-based firms competing against one another for 100,000 consumers over 10 periods.

We then evaluate whether optimized LLM agents outperform these heuristic firms under identical market conditions.

For each LLM model, we run five independent markets containing one optimized firm and five rule-based firms, each competing for 100,000 consumers over 10 periods. Figure 10 reports the resulting cumulative profits for the optimized agent and the mean of its rule-based competitors. Across all models, it was possible to assess that optimized LLM agents achieved substantially higher profits than rule-based firms. The performance gap grows monotonically with model capability: GPT-3.5-turbo yields a modest advantage, GPT-4o-mini achieves a large improvement, and GPT-5-mini delivers the strongest gains. To quantify this trend, we compute the Pearson correlation between each model’s average cumulative profit and its publicly reported MMLU(-Pro) score, obtaining

$$\rho = 0.9109,$$

indicating a strong positive association between LLM general capability and the market performance in the simulated market. While this does not imply causality, it provides evidence that tasks requiring strategic adaptation, economic reasoning, and multi-step inference benefit substantially from higher model sophistication, even when behavior is constrained to a simple, structured prompt interface. Also, it shows how more capable models are able to extract a higher values from the optimized prompts.

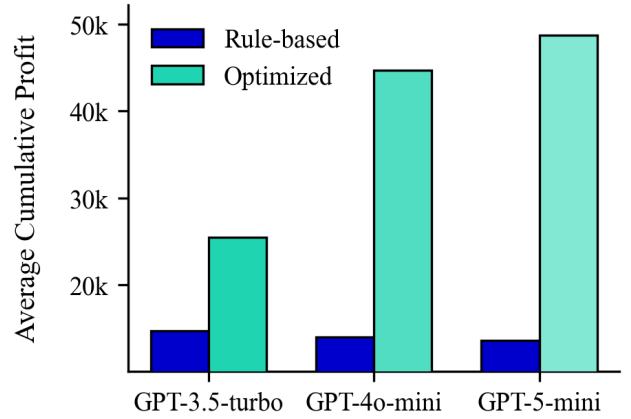


Fig. 10. Average cumulative profit of optimized LLM agents versus rule-based firms across three model families.

E. Cross-Model Robustness and Relationship to Model Capability

To evaluate whether the proposed evolutionary framework generalizes across architectures and model families, we replicated the full prompt-optimization pipeline using a diverse set of LLMs: GPT-3.5-turbo, GPT-4o-mini, GPT-5-mini, GPT-5-nano, Gemini-2.0-flash, Gemini-2.5-flash and Gemini-3-flash-preview. Across all models, the GA remained functional and consistently produced non-trivial improvements in cumulative profit, indicating that the method is robust to architectural differences, training paradigms, and variations in default sampling behavior.

Each model was evaluated over three independent GA runs to assess the stability of the convergence. For models with explicit temperature control (e.g., GPT-3.5-turbo, GPT-4o-mini, Gemini-2.0-flash, Gemini-2.5-flash, Gemini-3-flash-preview), the trajectories of mean active-bit density were tightly clustered, reflecting stable evolutionary dynamics and a relatively smooth fitness landscape. In contrast, GPT-5-mini and GPT-5-nano exhibited substantial divergence across runs. These models do not expose temperature control and therefore operate at a default sampling temperature of $T = 1$, which injects significant stochasticity into each decision-making step [54], [4]. The resulting run-to-run variability highlights the importance of controlling LLM sampling noise for reliable evolutionary optimization.

Across nearly all models, a common evolutionary pattern emerges. During early generations, the population displays an increase in active-bit density, indicating that the GA initially favors prompts containing a larger number of components. This inflation of genome density expands the exploratory capacity of the population and reduces the risk of early collapse in competitive markets. As evolution progresses, selection pressure systematically prunes redundant, neutral, and low-utility components, producing a characteristic decline in active-bit density. This emergent exploration–exploitation transition is visible in most trajectories reported in Figures 11 and 12, and is consistent with classical GA behavior in high-dimensional, noisy optimization landscapes [35].

Figure 11 illustrates the evolution of active-bit density for GPT models. A noteworthy trend emerges: models with higher general capability, as measured by publicly reported MMLU scores, tend to converge toward solutions with fewer active components (correlation between bit activation and MMLU = -0.96). This may suggest that more capable models require less explicit prompting structure to generate economic rational behavior, reflecting a form of information efficiency in their internal decision-making.

Interestingly, the trend observed for GPT models appears to be reversed for Gemini models (Fig. 12). In this case, higher model capability—as measured by MMLU(-Pro)—is associated with convergence toward prompt configurations with a greater number of active components. This divergence suggests that the relationship between model capability and optimal prompt complexity may be architecture-dependent, potentially reflecting differences in instruction adherence, reasoning granularity, or internal prompt integration mechanisms[55].

F. Limitations

While the proposed evolutionary framework demonstrates consistent and interpretable improvements in LLM-driven agent performance within the studied setting, several limitations should be acknowledged.

First, the prompt-optimization process operates over a discrete and pre-defined library of modular system prompt components. As a result, the search space is restricted to combinations of existing instructions rather than the full, unconstrained

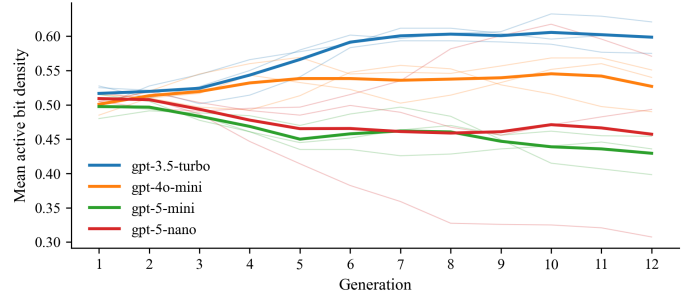


Fig. 11. Evolution of mean active-bit density across generations for GPT models. Thick lines represent the mean across three independent GA runs, while thin lines show individual runs.

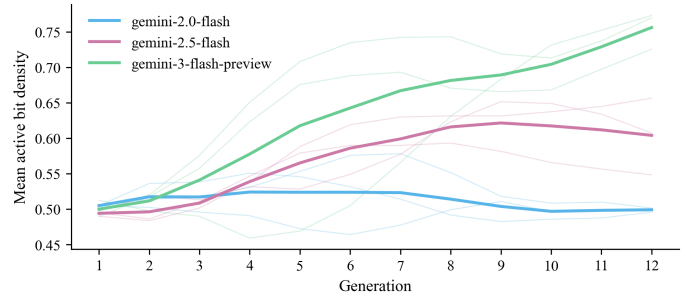


Fig. 12. Evolution of mean active-bit density across generations for Gemini models. Thick lines denote averages over three runs and thin lines denote individual runs.

space of possible prompt formulations. This design choice enables controlled analysis and post-hoc inspection of prompt utility, but does not capture richer degrees of freedom such as free-form rewriting, instruction ordering, stylistic variation, or exemplar-based prompting, all of which may further influence agent behavior.

Second, optimization in this study is restricted to the system prompt, while the user prompt is held fixed throughout all experiments. This separation isolates the effect of system-level instructions and substantially reduces search complexity; however, system and user prompts jointly shape model behavior in practice. Joint or hierarchical optimization of both channels may lead to different prompt structures and potentially stronger or more stable performance.

A further limitation arises from the co-evolutionary evaluation scheme adopted in this work. Because fitness is assessed through direct competition among contemporaneous prompt configurations, optimized prompts may partially specialize to the specific competitive dynamics and market design considered here. This path dependence may limit direct transfer to structurally different environments, such as markets with alternative utility specifications, demand models, or competitive rules. Still, it does not affect the overall relevance of the methodology, or its applicability.

In addition, evolutionary dynamics appear to be sensitive to interactions between GA hyperparameters and the underlying language model, even if it brings an advance over non-

optimized models with every model. In practice, hyperparameter settings that yielded stable evolutionary behavior for one model did not necessarily transfer to others, with some configurations leading to instability or collapse when applied across different LLMs. As a result, the unified configuration reported in this study was obtained through a trial-and-error calibration process aimed at identifying parameter values that enabled stable convergence across all evaluated models. Establishing principled, model-independent guidelines for hyperparameter selection in evolutionary prompt optimization therefore remains an open methodological challenge.

Finally, the analysis linking model capability to both economic performance and prompt sparsity relies on a limited set of language models and on MMLU(-Pro) scores as coarse proxies for general intelligence. While the observed correlations are suggestive, evaluating a larger and more diverse model suite and incorporating task-specific or behaviorally grounded capability measures would be necessary to strengthen the robustness and generality of these conclusions.

G. Results implications

The results of this study have several methodological and conceptual implications for the design and control of LLM-driven agents in interactive environments. Most notably, they suggest that prompt engineering for autonomous agents can be framed as a black-box optimization problem driven solely by realized task-level rewards, without requiring gradient information or access to internal model states. Although the genetic algorithm is not allowed to generate novel prompts, and is instead restricted to activating or deactivating a limited set of predefined modular components, this constraint has the purpose to exclude unconstrained textual modifications, stylistic reformulations, and variations in the ordering of instructions from the search space. This perspective highlights evolutionary prompt optimization as a practical approach even for complex settings characterized by delayed feedback, stochastic multi-agent interactions, and non-differentiable objectives, a set of problems that includes, among others, economic problems, negotiation tasks, and other social and computational complex systems domains.

A second implication concerns the role of prompt structure as a mechanism for behavioral control and post-hoc explainability analysis. By restricting optimization to modular prompt components, the evolutionary process yields sparse and stable prompt configurations whose functional contributions can be directly inspected. Also, the observed association between model capability and optimized prompt structure suggests that prompt complexity may act as an informative signal of how efficiently different LLMs internalize task constraints. So, in a way, the relationship between the specifics of the problem, the optimized prompt, and the model’s capability could be used to assess and better explain a model behaviour, especially in a complex and multi-entities setting.

V. CONCLUSIONS

This paper presented an evolutionary framework for optimizing system prompts of LLM-driven agents operating in

competitive, multi-agent market environments. By framing prompt design as a discrete, black-box optimization problem and applying a binary GA, we demonstrated the feasibility of developing effective agent behaviors without relying on manual instruction adjustments or model fine-tuning.

Through controlled agent-based market simulations, we showed that the proposed approach consistently converges to stable and structurally interpretable prompt configurations under controlled stochasticity. Firms equipped with optimized prompts achieve substantial economic advantages over both unoptimized LLM baselines and transparent rule-based competitors. We further observed that evolutionary convergence is sensitive to LLM-induced stochasticity, and that both economic performance and the structure of optimized prompts exhibit systematic variation across models with different capability levels.

Beyond performance gains, these results highlight prompt structure itself as a meaningful object of analysis. The emergence of sparse and stable prompt configurations, together with their association with model capability, suggests that prompt optimization may serve not only as a practical control mechanism, but also as an empirical technique for studying how LLM architectures employ their general competence to take decisions.

Several directions for future work naturally follow from these findings. First, the optimization framework could be extended beyond binary component selection to incorporate richer prompt transformations, such as controlled rewriting, instruction ordering, or the joint and hierarchical co-evolution of system and user prompts. Second, robustness could be improved by evaluating fitness across multiple stochastic market realizations or under domain randomization, thereby optimizing expected rather than realized performance and reducing sensitivity to particular competitive configurations. Third, the evolutionary objective could be generalized to multi-objective settings that jointly consider profit, risk, survival, or welfare metrics, enabling systematic exploration of trade-offs between economic performance and strategic stability.

Finally, extending the experimental scope to larger and more diverse model suites, longer time horizons, and more realistic market environments would allow stronger conclusions regarding generalization and transfer. In particular, dedicated studies integrating task-specific or behaviorally grounded capability measures alongside general benchmarks such as MMLU(-Pro) would be required to clarify which aspects of model competence are most relevant for strategic interaction. Together, these extensions would further establish evolutionary prompt optimization as a principled methodology for studying and deploying LLM-based agents in complex, competitive environments.

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